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Lensless camera positioned behind display layer

Abstract

Systems and methods are described for enabling a lensless camera having an image sensor and a mask to be positioned behind a display screen of a device, which allows for the device to have an increased screen-to-body ratio. The image sensor captures an image based on the light that travels through the display screen and the mask. The display screen may include portions between pixel elements that allow light to pass through. The mask may include a pattern, such as an opaque material with portions that allow light to pass through from the portions of the display layer to the image sensor. The image captured by the image sensor may be indiscernible to humans. The system may utilize a trained machine learning model to reconstruct the image, using data about the pattern of the mask, so humans may visually recognize features in the image.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) This disclosure is directed to systems and methods for using a lensless camera that is positioned underneath a display layer.

BACKGROUND

(2) Devices having display screens, such as smartphones, tablets, and laptops, may have a front-facing camera for taking selfies or making video calls. The front-facing camera may be installed on the same side of the device as its screen, such as the front side that faces a user. The front-facing camera may be installed in a top bezel or a notch cut-out of the screen. However, the bezel and notch cut-out require space that could otherwise be occupied by the screen, which limits the maximum screen-to-body ratio of the display screen. Front-facing cameras may also be installed at

a location that is away from the area of focus of the screen, such as the top or side of the screen. These locations draw the eye away from the screen, which may hinder video communication, where eye contact with a party on the screen may be desired. Thus, a device having a front-facing camera within the area of focus of the screen, without reducing the size of the screen, is needed.

(3) In one approach, the front-facing camera resides inside the device and rises to protrude from the device when needed. The camera may rise up automatically, such as when a user opens a camera app on the device, or may be manually raised by the user. While this approach may not reduce the screen size, it requires storage space inside the device, which may be needed to house other components and circuits of the device. This approach also requires a lifting mechanism to raise and lower the camera, which may increase design costs or introduce additional failure modes into the device. Further, the camera is positioned even further from the area of focus of the screen and may reduce eye contact during video calls.

(4) In another approach, the device has a rotatable rear-facing camera that may rotate to face the front of the device, becoming the front-facing camera. The rotatable camera may protrude from the device when functioning as the front-facing camera, or may be positioned in a notch of the front of the device. The rear-facing camera may rotate automatically when needed or manually by a user. This approach may have a limited impact on internal storage space since the rear-facing camera is not stored inside the device. However, this approach may require a rotational mechanism that may present similar challenges as the lifting mechanism of the front-facing camera previously discussed. This approach also draws the user's gaze away from the area of focus of the screen and does not improve eye contact during video calls.

(5) In another approach, the front-facing camera may be a miniature lens-based camera placed under the screen. While this approach may improve video communication by allowing the user to look at the party on the screen, it requires light to travel through the screen, through the lens of the camera, and to camera sensors of the miniature lens-based camera, which may result in low-light transmission and diffraction effects that negatively impact the quality of the captured images. The miniature lens-based camera may also have a small lens aperture and small sensor size that adversely impacts the quality of the captured image. A sophisticated computational model may be required to enhance the quality of the captured images. Thus, the miniature lens-based camera placed under the screen may not provide adequate image quality and may require powerful computational resources, which may limit the types of devices available to implement this approach.

(6) Further, the lens-based optical system of the miniature lens-based camera requires a distance between the lens and camera sensors, which may require an increased thickness of the device since the camera is positioned behind the screen. The distance between the lens and camera sensors may be reduced to compensate but may require a reduced sensor size that can adversely affect image quality.

(7) Accordingly, there is a need to provide a front-facing camera that provides a high-quality image and allows for an increased screen-to-body ratio. Such a solution leverages the thin, compact footprint of an image sensor and a mask to capture images.

SUMMARY

(8) To help address these problems, systems and methods are provided herein that enable a lensless camera having an image sensor and a mask to be positioned behind a display screen of a device, which allows for the device to have an increased screen-to-body ratio. A machine learning (ML) model is also provided herein to generate images based on the light that travels through the display screen and the mask and is received by the image sensor.

(9) In one approach, a lensless camera includes an image sensor (e.g., a complementary metal-oxide-semiconductor (CMOS) sensor) and a mask that are positioned behind a display layer of a device. The image sensor captures an image based on the light that travels through the display layer and the mask. Thus, in some embodiments, the lensless camera may include the image sensor, the

mask, a portion of the display layer disposed above the mask (e.g., the portion that allows light to travel through to the mask). The captured image may be indiscernible to humans. The system may utilize a computing system that is communicatively connected to the image sensor for capturing an image of an environment or a feature (e.g., by converting detected photons into electric signals). The display layer may include portions between pixel elements that are at least partially transparent (referred to as portions of the display layer), where the portions of the display layer allow light to pass through. The image sensor may include a physical mask that comprises a predetermined pattern (e.g., a pinhole pattern). For example, the physical mask may comprise an opaque material (e.g., a fully non-transparent material) with portions that are at least partially transparent (e.g., that allow light to pass through from the portions of the display layer). The portions of the mask that are at least partially transparent (referred to as portions of the mask) may be shaped according to the pattern. Alternatively, the pattern may refer to the pattern of the opaque material. The mask pattern may be generated using random spacing of opaque or clear bands and/or shapes, or created according to an obfuscated algorithm, or manually created by a provider or by a user. The pattern of the mask may distort (e.g., blur) the captured image data, for example by casting a shadow on the light-sensing part of the camera and/or by causing refraction in the incoming light.

(10) Because the camera may be lensless and may not focus the incoming light the computing system may not be able to reconstruct the distorted image of the environment based only on the light that passed through the pattern. Giving the computing system access to data that represents the pattern of the mask and layout of the portions of the display layer allows the computing system to reconstruct the image of the environment. The computing system may use a trained ML model to generate the reconstructed image.

(11) In one approach, the device prevents light emitted by the display layer from reaching the image sensor. In some embodiments, the device may use a casing around the pixel elements to block the light from reaching the image sensor, or to direct the light away from the image sensor. In some embodiments, the device may also ensure the pixel elements near the image sensor do not emit light when the image is captured.

(12) In some embodiments, the ML model is trained using input and output training images. The ML model uses the input training image and data about the pattern of the mask to generate an attempted reconstructed image, which is then compared to the output training image. The ML model is updated based on the comparison and another attempted reconstructed image is generated. This process is repeated until the attempted reconstructed image is sufficiently close to the output training image. In some embodiments, the ML model is trained using large datasets or batches of input and output training image pairs. The training process may be repeated until the attempted reconstructed images are sufficiently close to the output training images.

(13) In some embodiments, the data about the pattern of the mask may be an image (or series of images) that is based on light that passed through the mask. In some embodiments, the training input image contains the data about the pattern of the mask. In some embodiments, the data about the pattern of the mask further includes data about the portions of the display layer, since the light that passed through the mask may have previously passed through the display layer. In some embodiments, the data about the pattern of the mask may be separately provided into the ML model. In some embodiments, the ML model may be trained for one specific pattern of the mask. In such embodiments, multiple masks may be produced having the same specific pattern and each of the masks may be used in a similar, but physically different device (e.g., a model of a smartphone).

(14) In some embodiments, the training output image contains depth information relating to distances of features contained therein. For example, if the output training image was taken by a camera, then the depth information may contain distances from the camera to the features in the output training image. In some embodiments, an array of cameras may capture images of the features at slightly different angles or orientations. In such embodiments, the output training image

may be computed from the images taken by the camera array. The training input image may contain depth information. The image sensor may capture light at different angles, e.g., light that has reflected off features in different positions of the environment. The ML model may apply attempted reconstructed depth information to the attempted reconstructed image to create an output image and compare the output image to the output training image. The ML model is updated until the attempted reconstructed depth information is sufficiently close to the depth information of the training output image. The ML model may be trained to generate the reconstructed depth information simultaneously while being trained to generate the reconstructed image, or in separate training operations. In some embodiments, the output image is a depth image.

(15) In some embodiments, the pattern of the mask may be configurable. For example, the system may configure opaque portions of the mask to change how light is blocked. The configurable pattern may increase privacy. For example, if the pattern of the mask is changed from an initial pattern on which the ML model was trained, the ML model may not be able to generate the reconstructed image using the captured image without retraining. Thus, the environment, features, or people in the captured images remain indiscernible in the reconstructed images until the pattern is changed back to the initial pattern. In some embodiments, the entire mask may be set to opaque when the image sensor is not used.

(16) Using the methods described herein, an image sensor having a mask may be positioned under a display screen. This approach allows a device having the display screen to maximize a screen-to-body ratio and have a thickness that is less than if using a lens-based camera. The lensless, under-display cameras may also improve video communication, allowing the camera to be placed in-line with the eyes of the person in the video call, making eye contact feel more natural. Multiple under-display cameras may be used at different locations under the display to make the gaze adjustment much easier than having a camera array located outside the display screen.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure, in accordance with one or more various embodiments, is described in detail with reference to the following figures. The drawings are provided for purposes of illustration only and merely depict typical or example embodiments. These drawings are provided to facilitate an understanding of the concepts disclosed herein and should not be considered limiting of the breadth, scope, or applicability of these concepts. It should be noted that for clarity and ease of illustration, these drawings are not necessarily made to scale.

(2) FIG. 1 shows an illustrative diagram of an under-display camera, in accordance with some embodiments of this disclosure;

(3) FIG. 2 shows a side-sectional view of a display layer, in accordance with some embodiments of this disclosure;

(4) FIG. 3 shows a side-sectional view of a display layer, in accordance with some embodiments of this disclosure;

(5) FIG. 4 is a flowchart of a detailed illustrative process for generating a reconstructed image, in accordance with some embodiments of this disclosure;

(6) FIG. 5A is a flowchart of a detailed illustrative process for training a machine learning model for image reconstruction, in accordance with some embodiments of this disclosure;

(7) FIG. 5B shows an illustrative technique for training a machine learning model for image reconstruction, in accordance with some embodiments of this disclosure;

(8) FIG. 6 shows an illustrative user equipment device, in accordance with some embodiments of this disclosure;

(9) FIG. 7 shows an example system, in accordance with some embodiments of this disclosure;

(10) FIGS. 8A-8C show illustrative diagrams of adjusting a pattern of a mask, in accordance with some embodiments of this disclosure; and

(11) FIGS. 9A and 9B show illustrative diagrams of arrays of pixel elements, in accordance with some embodiments of this disclosure.

DETAILED DESCRIPTION

(12) FIG. 1 shows an illustrative diagram of an under-display, lensless camera **100**, in accordance with some embodiments of this disclosure. The lensless camera **100** is part of a device **101**, which is depicted as a phone. In some embodiments, the device may be a television, computer monitor, video game system, or any device having a screen. The lensless camera **100** includes an image sensor **108** that is positioned under a display layer **102** of the device **101**. The lensless camera **100** further includes a mask **110** that is disposed between the image sensor **108** and the display layer **102**.

(13) Light reflects off an object or feature **130**, shown as a dog, and travels through the display layer **102**, through the mask **110**, and to the image sensor **108**, which captures a captured image **140** of the feature **130**. The captured image **140** may be visually indecipherable to humans. Control circuitry **142** inputs the captured image **140** into a trained machine learning (ML) model **144**. The ML model **144** generates a reconstructed image **146** of the feature **130** that humans can visually recognize. The control circuitry **142** retrieves the reconstructed image **146** from ML model **144** and stores it in a memory **148** of the device **101**. The ML model **144** may reside in the memory **148**. In some embodiments, the ML model **144** may place the reconstructed image **146** in the memory **148**, e.g., without the control circuitry **142**.

(14) The image sensor **108** may include a complementary metal-oxide-semiconductor (CMOS) sensor, a charge-coupled device (CCD) sensor, or any other suitable type of a light sensor configured to produce pixel data that is based on light hitting a light-sensitive front of the image sensor **108**.

(15) The mask **110** may be manufactured out of any suitable opaque material **114** that prevents penetration by light. For example, mask **110** may be manufactured using plastic, metal, wood, (stained) glass or any other suitable opaque materials. The mask **110** comprises portions **112** that are at least partially transparent, forming a pattern. The portions **112** may allow light that penetrates the display layer **102** to further penetrate the mask **110** and travel to the image sensor **108**. The pattern of the portions **112** (also referred as the mask pattern or the pattern of the mask **110**) may be defined randomly. For example, the pattern of the portions **112** may include horizontal and vertical cuts of different lengths and widths. In another example, the horizontal and vertical cuts may be randomly spaced and/or be a randomly selected length and width. In another example, a certain number of holes may be selected. The holes may have different shapes, such as rounded, circular, or polygonal shapes, that have a known or random position.

(16) The pattern may also be created using an algorithm, such as a hidden algorithm not accessible to anyone trying to decode the pattern. The pattern may also be selected by a manufacturer or by the end user. For example, the user may be able to define any kind of pattern and mask **110**, which may be a three dimensional (3D) printed mask, with the defined pattern. The term “pattern” may also refer to the configuration of the opaque material **114** of the mask **110**.

(17) In some embodiments, the portions **112** of mask **110** may contain or be filled with a transparent material (e.g., glass or clear plastic) or semi-transparent material. In some embodiments, the mask **110** may include a transparent material (e.g., glass or clear plastic) with a pattern of opaque markings printed on the transparent material. In some embodiments, the transparency of the mask **110** may be configurable such that the shape and area of the portions **112** and the opaque material **114** may be changed.

(18) The mask **110** may be positioned over the image sensor **108** during the manufacturing process, or in post-production. In some embodiments, the mask **110** may be sold or provided separately from image sensor **108** and positioned over the image sensor **108** by an end user or installation

technician, such as through a slot in the device **101** (e.g., similar to a memory card or subscriber identity module (SIM) card).

(19) The display layer **102** may include any suitable display technology that allows light to pass through at least when the display is off. In the depicted embodiment, a portion of the display layer **102** positioned above the image sensor **108** is shown and a remaining portion of the display layer **102** is omitted to avoid overcomplicating the drawing. The display layer **102** includes a plurality of pixel elements **104** and portions **106** between the pixel elements **104** that are at least partially transparent. The pixel elements **104** emit light of different colors and are shown in a grid pattern with the corners touching another. Each pixel element **104** comprises three subpixels **120** (e.g., red, green, and blue). In some embodiments, the pixel element **104** may include more or fewer subpixels **120** (e.g., red, green, blue, and white). In some embodiments, the subpixels **120** may each be a different size or have a different area. Some embodiments may include pixel elements **104** having more subpixels **120** that emit a first color (e.g., green) than subpixels **120** that emit a second color and third color (e.g., blue and red). In some embodiments, the pixel elements **104** may not have subpixels **120** and each pixel element **104** may emit light of a color (e.g., red, green, or blue). The portions **106** of the display layer **102** may include any of openings, lenses, transparent material, semi-transparent material, or any configuration that allows light to at least partially pass through, e.g., to the mask **110** or to the image sensor **108**.

(20) Once assembled, the display layer **102** and the mask **110** may form a coded aperture of the lensless camera **100**. The display layer **102** and the mask **110** may each block or refract some of the light from traveling to image sensor **108** that is reflected off the feature **130**. The mask pattern allows light that is directed through, passes through, or otherwise travels through a first subset of the portions **106** to be directed to the image sensor **108**, since the mask **110** is positioned under the display layer **102**. The opaque material **114** of the mask **110** blocks light such that the image sensor **108** of lensless camera **100** (e.g., CMOS sensor, CCD sensor, etc.) is able to receive light only through the first subset of the portions **106** of the display layer **102**, and through the portions **112** in the mask **110** to form heavily distorted (e.g., blurred) image data (e.g., the captured image **140**). For example, light that is directed through a second subset of the portions **106** may be blocked from reaching the image sensor **108** by the opaque material **114**. Thus, the captured image **140** may contain data defining the pattern of the mask **110** and a layout (or pattern) of the portions **106** of the display layer **102**. The second subset of the portions **106** may be different than the first subset of the portions **106**. In some embodiments, the second subset of portions **106** may include portions **106** that are not in the first subset of portions **106**. In some embodiments, the second subset of portions **106** is positioned entirely above the mask **110**.

(21) The control circuitry **142** may generate the captured image **140** based on the light that is directed through the first subset of the portions **106**, through the pattern of the mask **110**, and to the image sensor **108**. In some embodiments, the image sensor **108** may generate the captured image **140** and the control circuitry **142** may interface with the image sensor **108** (or the lensless camera **100**) to retrieve the captured image **140**. As will be explained, an undistorted image (e.g., training output image **545** in FIG. 5B) is not recoverable based on pixel data (e.g., the captured image **140**) produced by the image sensor **108** without data that defines at least the dimensions of the pattern of mask **110**. In some embodiments, additional data that defines the properties of the mask **110** may be needed (e.g., sensor field of the mask **110**).

(22) The control circuitry **142** generates the reconstructed image **146** based on the captured image **140** and the data defining the pattern of the mask **110**. For example, the control circuitry **142** inputs the captured image **140** into the ML model **144**, which was captured using light that was directed through the portions **106** of the display layer **102** and the pattern of the mask **110**, and outputs the reconstructed image **146** to the control circuitry **142**. The ML model **144** may access the memory **148** when creating the reconstructed image **146**. The reconstructed image **146** is stored in the memory **148** and may be displayed on the display layer **102**. In the depicted embodiment, the

control circuitry **142** is part of the device **101**. In some embodiments, the control circuitry **142** may be external to the device **101** and may reside on, e.g., a server or cloud computing device.

(23) In some embodiments, the control circuitry **142** changes the pattern of the mask **110** by turning on and off the transparency of the mask **110** or by adjusting the opacity of the mask **110**. In some embodiments, the control circuitry **142** may change the shape of the portions **112** (e.g., transparent material) or the shape of the opaque material **114**. For example, the control circuitry **142** may configure the mask **110** to be opaque or fully opaque when the image sensor **108** is not active (e.g., not powered or capturing images). In some embodiments, such as when the mask **110** is opaque, the opaque material **114** of the mask **110** is configured to block light directed through the first subset of the portions **106** of the display layer **102** from traveling to the image sensor **108**. In some embodiments, such as when the mask **110** is fully opaque, the transparency of the mask **110** is turned off. For example, the mask may be configured such that the opaque material **114** covers the entire surface of the mask **110** facing the display layer **102** and does not allow light to travel through the mask **110**. The control circuitry **142** may set the mask to be opaque or fully opaque when a subset of the pixel elements **104** emit light and the image sensor **108** is not active. The subset of the pixel elements **104** emitting light may be disposed above the mask **110** or above portions of the mask **110** disposed over the image sensor **108**. The control circuitry **142** may set at least a portion of the mask **110** to be transparent when the subset of the pixel elements **104** do not emit light and the image sensor **108** is active. In some embodiments, the control circuitry **142** may control the pixel elements **14** and may alternate between the subset of the pixel elements **104** emitting light and the image sensor **108** being active at a predetermined frequency. In some embodiments, the predetermined frequency is a frequency that is ergonomic to the human eye. The predetermined frequency may be a frequency where the human eye does not notice a flicker in the display layer **102**. In some embodiments, the predetermined frequency may be at least 24 Hz. In some embodiments, the mask **110** includes an electronic ink layer, e.g., for the opaque material **114**, and the control circuitry **142** reconfigures the electronic ink layer to turn on and off the transparency of the mask **110**. In some embodiments, the electronic ink layer may be a substantially similar size as the display layer, such as within an area of 99%, such as within 95%, such as within 90%. In some embodiments, the electronic ink layer may be used to display content (e.g., an image, text, and the like) to a user of the device **101**. The electronic ink layer may be used instead of the display layer **102**, such as when the display layer is turned off (e.g., the pixel elements **104** do not emit light).

(24) In some embodiments, the display layer **102** includes a bottom substrate layer, a cathode layer, an anode layer, and an emission layer. In such embodiments, the bottom substrate layer may include a transparent or semi-transparent material, the cathode layer may inject electrons, and the anode layer may receive the injected electrons. The emission layer may be positioned in between the cathode and anode layers, and may emit light when the ejected electrons pass through. The emissive layer may include organic plastic molecules, and a color of the emitted light may depend on the type of the organic molecules. The intensity of the emitted light may depend on a magnitude of a current applied to inject the electrons. In some embodiments where the substrate layer is transparent, the other layers of the display layer **102** may also be transparent such that the display layer **102** is transparent when turned off and light may be directed through the display layer **102** from either direction. In some embodiments, the mask **110** is part of the display layer **102**. For example, the mask **110** may be printed on, bonded to, or formed within a the bottom substrate layer of the display layer **102**.

(25) In some embodiments, transparency of the portions **106** may be controlled, e.g., by the control circuitry **142**, similar to the transparency of the mask **110**. In some embodiments, the display layer **102** includes a liquid crystal display (LCD) and its derivative technologies, such as light-emitting diode (LED), quantum dot LED (QLED), mini-LED, and micro-LED, to name a few examples. In some embodiments, the display layer **102** may comprise an organic light-emitting diode (OLED).

The display layer **102** may display an image that is different than the captured image **140** and the reconstructed image **146**, such as a media asset **103**, which is shown as a social media post. Thus, a user may view the media asset **103** on the display layer **102** that is unrelated to operation of the image sensor **108**. In some embodiments, the display layer **102** may display the media asset **103** while the image sensor **108** captures the captured image **140**.

(26) In some embodiments, the mask **110** may be an amplitude-coded mask. In some embodiments, the mask **110** may be a phase mask. In some embodiments, the mask **110** may be one of a diffuser, diffraction grating, or Fresnel lens plate, to name a few examples.

(27) In some embodiments, multiple image sensors **180** and masks **110** may be placed under the display screen **102** to form multiple lensless cameras **100** of the device **101**. The multiple lensless cameras **100** may be used to align the reconstructed image **146** with a user's eye, such as aligning with a pupil of the eye, so the user appears to maintain eye contact with a viewer of the reconstructed image **146**. The multiple lensless cameras **100** may be used to capture captured images **140** at different angles or orientations, which may help generate a depth map as discussed in FIG. 5B. The multiple lensless cameras **100** may be used to simultaneously capture multiple captured images **140**, which may be used to compensate for a low amount of light that traveled the image sensors **108**. Such an embodiment may be used for capturing images where ambient light is low, such as when the ambient light is at or below **180** lumens, such as at or below **50** lumens, such as at or below **15** lumens. The multiple lensless cameras **100** may be used to simultaneously capture multiple captured images **140** that may be “stitched” together (e.g., by the control circuitry **142**) to create one image with a larger resolution and/or field-of-view.

(28) FIG. 2 shows a side-sectional view of a display layer **202**, in accordance with some embodiments of this disclosure. The display layer **202** includes pixel elements **204** disposed on a transparent substrate, and portions **206** between the pixel elements **204** that are at least partially transparent. Each pixel element **204** includes subpixels **220** that emit different colors (e.g., red, green, and blue). The portions **206** are depicted as openings between the subpixels **220** and portions of the transparent substrate in-line with the openings. The mask **110** and the image sensor **108** are disposed below the display layer **202** such that light is directed through a subset of the portions **206**, through the pattern of the mask **110**, and to the image sensor **108**.

(29) The subpixels **220** each include a casing **222** to block light emitted by the subpixels **220** from traveling through the portions **206** and to the image sensor **108**. The casing **222** may include an opaque material. The subpixels **220** may be arranged in different patterns. For example, a pattern of the subpixels **220** may be such that the pixel elements **204** do not form a grid pattern with the corners of the pixel elements **204** touching another, such as described in relation to FIG. 1. In some embodiments, the pattern of the subpixels **220** may be equally spaced. In some embodiments, the configuration of the colors emitted by the subpixels **220** may be arranged differently. In some embodiments, the subpixels **220** of each pixel element **204** may emit other or additional colors, such as white.

(30) In some embodiments, the casing **222** may cover a side of the subpixels **220** that faces the image sensor **108**. In some embodiments, the casing **222** surrounds all but one side of the subpixels **220** to only allow light emission in a direction facing away from the image sensor **108**. In some embodiments, the casing **222** may cover the backside of the subpixels **220** that face the image sensor **108** and a portion of sidewalls of the subpixels **220** that face the portions **206**.

(31) In some embodiments, the casing **222** may be disposed around the pixel elements **204**, which includes the subpixels **220**. For example, the casing **222** may extend between the subpixels **220** such that the casing **222** blocks light from being directed through the portions **206** between the subpixels **220**. In some embodiments, the pixel elements **104** discussed in relation to FIG. 1 may comprise the casing **222**.

(32) In some embodiments, the pixel elements **204** include stacked subpixels **220**. For example, the pixel elements **204** may include a subpixel **220** stack having emission layers (e.g., one for each

color emitted) and electrodes (e.g., transparent electrodes) between the emission layers and on top of the uppermost emission layer. An electrode (e.g., transparent, opaque, or semi-transparent electrode) may be between the pixel stack and a substrate (e.g., transparent, opaque, or semi-transparent substrate) of the display layer **202**. The stacked subpixel **220** configuration may allow for the portions **206** of the display layer **202** to be larger, which may direct more light through the display layer **202**.

(33) FIG. **3** shows a side-sectional view of a display layer **302**, in accordance with some embodiments of this disclosure. The display layer **302** includes the pixel elements **204** disposed on a substrate, which may or may not be transparent. The substrate comprises a plurality of microlenses **324**. The microlenses **324** are disposed in the substrate and are aligned with portions **306** of the display layer **102** that are at least partially transparent. The portions **306** are depicted as openings between the subpixels **220**. The subpixels **220** include the casings **222**. The mask **110** and the image sensor **108** are disposed below the display layer **202** such that light is directed through a subset of the portions **306**, through a subset of the microlenses **324**, through the pattern of the mask **110**, and to the image sensor **108**. The microlenses **324** may help focus and concentrate light that travels through the portions **306**, and then through the mask **110**, onto the image sensor **108**. Focusing the light may result in more photons of the light reaching the image sensor **108**, which may increase the brightness of the captured image **140** (FIG. **1**).

(34) In some embodiments, each microlens **324** may be disposed under a pixel element **204** such that each microlens **324** is disposed under all of the subpixels **220** of the pixel element **204**. In some embodiments, the microlenses **324** may be disposed between the pixel elements **204**, but not between the subpixels **220** of the pixel elements **204**. In some embodiments, the microlenses **324** may be disposed in a portion of the display layer **302** above the image sensor **108** such that the microlenses **324** are not disposed in other portions of the display layer **302** (e.g., portions not above the image sensor **108**).

(35) In some embodiments, the microlenses **324** may be disposed between the portions **306** of the display layer **302** and the portions **112** (FIG. **1**) of the mask **110** such that a pattern of the microlenses **324** is “coded” to or based on the pattern of the mask **110**. For example, the microlenses **324** may not be present in the portions **306** above the opaque material **114** (FIG. **1**) of the mask **110**. In some embodiments, the microlenses **324** may be disposed in a microlens layer that is separate than the display layer **302**. In such embodiments, the portions **306** may include openings and/or transparent or semi-transparent material. The microlenses **324** of the microlens layer may be aligned with the portions **306** of the display layer **302**. In some embodiments, the microlenses **324** may be disposed in the mask **110**, such as in the portions **112** of the mask **110**. In some embodiments, the display layer **302** and the microlenses **324** may be used without the mask **110**.

(36) FIG. **4** is a flowchart of a detailed illustrative process **400** for generating a reconstructed image (e.g., the reconstructed image **146** in FIG. **1**), in accordance with some embodiments of this disclosure. The process and operations shown in FIG. **4** may be implemented, in whole or in part, by one or more systems or devices described herein.

(37) The process **400** includes operation **402** with accessing, e.g., using control circuitry (e.g., control circuitry **142** in FIG. **1** and control circuitry **604**, **711** described below in FIGS. **6** and **7**), an image sensor (e.g., image sensor **108** in FIGS. **1**, **2**, and **3**) of a lensless camera (e.g., lensless camera **100** in FIG. **1** and described below in FIG. **5**) with a display layer (e.g., display layer **102**, **202**, **302** in FIGS. **1**, **2**, and **3** and display layer **902** and **962** below in FIGS. **9A** and **9B**) and a mask (e.g., mask **110** in FIGS. **1**, **2**, and **3** and mask **810** below in FIGS. **8A-8C**) that partially obscures the image sensor, such as described above with respect to FIG. **1**. The process optionally continues to operation **404** with turning off, e.g., using the control circuitry, pixel elements (e.g., pixel elements **104**, **204** in FIGS. **1**, **2**, **3**, and **8** and pixel elements **904** and **964** below in FIGS. **9A** and **9B**) of the display layer disposed above the mask. The process optionally continues to operation

406 with configuring, e.g., using the control circuitry, the mask to be partially transparent, such as described above with respect to FIG. 1. The process **400** continues to operation **408** with capturing an image (e.g., captured image **140** in FIG. 1) using the lensless camera. Operations **402**, **404**, **406**, and **408** may be performed before, after, or in parallel with operations **410**, **412**, and **414**. Operations **404** and **406** are optional because the image may be captured without turning off the pixel elements or configuring the mask.

(38) The process **400** continues to operation **410** with accessing, e.g., using the control circuitry, training input images of a feature, such as described below with respect to FIGS. 5A and 5B. The process **400** continues to operation **412** with accessing training output images of a feature, such as described below with respect to FIGS. 5A and 5B. The process **400** continues to operation **414** with training, e.g., using the control circuitry, an ML model (e.g., ML model **144** in FIG. 1 and below in FIG. 5B) to generate reconstructed images (e.g., reconstructed image **146** in FIG. 1 and below in FIG. 5B), such as described below with respect to FIGS. 5A and 5B. Operations **408** and **414** may be inputs into operation **416**. Each of operations **408** and **414** may occur at different times and at different frequencies than the others.

(39) The process **400** continues to operation **416** with inputting, e.g., using the control circuitry, the captured image into the ML model trained with regard to a pattern of the mask, such as described above with respect to FIG. 1. The process **400** continues to operation **418** with processing the captured image using the ML model to generate the reconstructed image. Thus, the ML model may generate the reconstructed image based on the captured image and data defining the pattern of the mask. The data defining the pattern of the mask may account for, or may include data defining a layout of portions of the display layer that are at least partially transparent (e.g., portions **106**, **206**, **306** in FIGS. 1, 2, 3, and 8 and portions **906** and **966** described below in FIGS. 9A and 9B), such as described above with respect to FIG. 1. The process **400** continues to operation **420** with a decision on whether to continue to capture images. If additional images are to be captured, the process returns to operation **408**. If no additional images are to be captured, the process **400** may optionally proceed to operations **422** and **424**, after which, the process **400** may end. In operation **422**, the process **400** optionally continues with turning on, e.g., using the control circuitry, the pixel elements of the display layer disposed above the mask. In operation **424**, the process **400** optionally continues with configuring, e.g., using the control circuitry, the mask to be fully opaque, such as described above with respect to FIG. 1.

(40) FIG. 5A is a flowchart of a detailed illustrative process **500** for training an ML model for image reconstruction, in accordance with some embodiments of this disclosure. The process and operations shown in FIG. 5A may be implemented, in whole or in part, by one or more systems or devices described herein.

(41) The process **500** optionally includes operation **502** with accessing, e.g., using control circuitry (e.g., control circuitry **142** in FIG. 1 and control circuitry **604**, **711** described below in FIGS. 6 and 7), an image sensor (e.g., image sensor **108** in FIGS. 1, 2, and 3) of a lensless camera (e.g., lensless camera **100** in FIG. 1 and described below in FIG. 5) with a display layer (e.g., display layer **102**, **202**, **302** in FIGS. 1, 2, and 3 and display layer **902** and **962** below in FIGS. 9A and 9B) and a mask (e.g., mask **110** in FIGS. 1, 2, and 3 and mask **810** below in FIGS. 8A-8C) that partially obscures the image sensor, such as described above with respect to FIG. 1. The process **500** optionally continues to operation **504** with capturing a training input image of a feature using the lensless camera, such as described below with respect to FIG. 5B. The training input image is captured using the light that is directed through the display layer (e.g., portions **106**, **206**, **306** in FIGS. 1, 2, 3, and 8 and portions **906** and **966** described below in FIGS. 9A and 9B), through the mask (e.g., pattern of the mask), and to the lensless camera. Thus, the training input image includes data defining the pattern of the mask, which may include data defining a layout of the portions of the display layer. The process continues to operation **506** with generating an attempted reconstructed image (e.g., reconstructed image **146** in FIG. 1 and below in FIG. 5B) using a generator (e.g.,

generator **150** described below in FIG. 5B) of an ML model (e.g., ML model **144** in FIG. 1 and below in FIG. 5B) and a training input image, such as described below with respect to FIG. 5B. Operations **502**, **504**, and **506** may be performed before, after, or in parallel with operations **508** and **510**.

(42) The process **500** optionally includes operation **508** with accessing, e.g., using the control circuitry, a reference camera (e.g., camera array **538** described below in FIG. 5B), such as described below in relation to FIG. 5B. The process **500** optionally continues to operation **510** with capturing a training output image (e.g., training output image **545** in FIG. 5B) of a feature using the reference camera (e.g., camera array **538** described below in FIG. 5B), such as described below in relation to FIG. 5B. Operations **506** and **510** may be inputs into operation **512**. Each of operations **506** and **510** may occur at different times and at different frequencies than another. Operations **504**, **506**, **508**, and **510** are optional because the training input image and/or the training output image may exist, e.g., as part of an image library or a database, such as described below with respect to FIG. 5B.

(43) The process **500** continues to operation **512** with comparing, e.g., using the control circuitry, an attempted reconstructed image (e.g., attempted reconstructed image **546** in FIG. 5B) to the training output image using a discriminator (e.g., discriminator **152** described below in FIG. 5B) and constructing a loss function, such as described below with respect to FIG. 5B. The loss function may calculate the differences between the attempted reconstructed image and the training output image.

(44) The process **500** continues to operation **514** with a decision on whether the attempted reconstructed image is adequate. If the attempted reconstructed image is not adequate, the process continues to operation **516** with adjusting, e.g., using the control circuitry, the generator based on the loss function, e.g., through backpropagation, and thereafter returns to operation **506**. Thus, the ML model may be trained using data defining the pattern of the mask. If the attempted reconstructed image is adequate, the process continues to operation **518** where the training is complete.

(45) FIG. 5B shows an illustrative technique for training the ML model **144** for image reconstruction, in accordance with some embodiments of this disclosure. In the depicted embodiment, the ML model **144** is trained using a dataset generated by the display layer **102**, mask **110**, and image sensor **108**, and a camera array **538**. Light reflected off the feature **130** travels to a beam splitter **532**, which splits the light into a first beam of light **534** and a second beam of light **536**. The first beam of light **534** travels to the lensless camera **100**. Only a portion of the first beam of light **534** travels through the display layer **102**, through the mask **110**, and to the image sensor **108**, such as described in relation to FIG. 1. The image sensor **108** generates or captures the training input image **540** based on the portion of the first beam of light **534**. Thus, the training input image **540** is specific to the pattern of the mask **110** (e.g., the opaque material **114** and portions **112** in FIG. 1). The image sensor **108** may use the mask **110** to capture light field information since the mask **110** directs light to the image sensor **108** from different viewpoints. For example, the mask **110** may be a phase mask that bends the portion of the first beam of light **534** to allow the image sensor **108** to capture the intensity and phase of the portion of the first beam of light **534**. The captured phase may be used by the ML model **144** to generate a depth map of the feature **130** in the training input image **540** (or in the captured image **140** in FIG. 1).

(46) The second beam of light travels to the camera array **538**, which is shown as having three cameras. Each camera of the camera array **538** is used to capture perspective images **539**, which are images that include the feature **130** taken from different angles. One of the three cameras may be positioned to have a substantially similar perspective as the lensless camera **100** and may capture an aligned image **541**. The control circuitry **142** may generate a training depth map of the feature **130** based on the perspective images **539**, e.g., by using stereo vision or stereo imaging techniques. The control circuitry **142** applies the training depth map to the aligned image **541** to generate the

training output image **545**. Thus, the training output image **545** includes a depth map (e.g., the training depth map). In some embodiments, the camera array **538** may have two cameras. In some embodiments, the camera array **538** may have four or more cameras.

(47) In the depicted embodiments, the ML model **144** includes a deep learning neural network, such as a general adversarial network (GAN). The ML model **144** comprises a generator **150** and a discriminator **152**. The ML model **144** is trained by accessing the training input image **540** and the training output image **545**. The training input image **540** is inputted into the generator **150**, which generates the attempted reconstructed image **546** and an attempted reconstructed depth map based on the training input image **540**. The discriminator **152** compares the attempted reconstructed image **546** and the attempted reconstructed depth map with the training output image **545** and the training depth map to construct a loss function. The loss function may include any of a reconstruction loss, a perceptual loss, and an adversary loss. The ML model **144** is adjusted based on the comparison. The generator **150** uses the loss function as feedback to improve its own performance, such as by updating its own parameters, and regenerates the attempted reconstructed image **546**. The discriminator **152** compares the regenerated attempted reconstructed image **546** to the training output image **545** and updates the loss function. This process is repeated until the attempted reconstructed image **546** adequately matches the training output image **545**, which is when the ML model **144** may be considered a trained ML model **144**.

(48) In some embodiments, training may be complete when the performance of the generator **150** and the discriminator **152** has stabilized. In some embodiments, human evaluation, such as through visual inspection of the attempted reconstructed image **546** and the training output image **545**, may be used to determine if training is complete. In some embodiments, a validation set of input and output images that differ from the training input image **540** and the training output image **545** (e.g., contain a different feature) may be used with the trained ML model **144** to verify training is complete. The resulting attempted reconstructed image may be evaluated using metrics such as the Fréchet Inception Distance or Inception Score, which provide a quantitative measure of the quality of the generated reconstructed image.

(49) In some embodiments, the ML model **144** is a U-Net GAN. The U-Net GAN may use a segmentation network as the discriminator **152**. In some embodiments, U-Net can be used as the generator **150**.

(50) In some embodiments, the ML model **144** trains to generate the attempted reconstructed image **546** and the attempted reconstructed depth map in separate processes. In some embodiments, the ML model **144** does not generate the attempted reconstructed depth map. In such embodiments, the mask **110** may not be a phase mask. In some embodiments, a single camera may be used instead of the camera array **538**. The single camera may have a substantially similar perspective to the lensless camera **100** and may be used to capture a single training output image, which may be used as the training output image **545**. During training of the ML model **144**, the discriminator compares the attempted reconstructed image **546** with the single training output image to generate the loss function.

(51) In some embodiments, the ML model **144** may use a training input image **540** and training output image **545** from a library of images or a database.

(52) FIGS. **6** and **7** depict illustrative devices, systems, servers, and related hardware for under-display image capture and image reconstruction. FIG. **6** shows generalized embodiments of illustrative user equipment devices **600** and **601**, which may correspond to, e.g., control circuitry **142**. For example, user equipment device **600** (referred to as device **600**) may be a smartphone device, a tablet, a virtual reality or augmented reality device, or any other suitable device capable of processing image or video data. In another example, device **601** (referred to as device **601**) may be a user television equipment system or device. Device **601**, which may be a television, may include set-top box **615**. Set-top box **615** may be communicatively connected to microphone **616**, audio output equipment **614** (e.g., speaker or headphones), and display **612**. In some embodiments,

display **612** may be a television display or a computer display. In some embodiments, set-top box **615** may be communicatively connected to user input interface **610**. In some embodiments, user input interface **610** may be a remote-control device. Set-top box **615** may include one or more circuit boards. In some embodiments, the circuit boards may include control circuitry, processing circuitry, and storage (e.g., RAM, ROM, hard disk, removable disk, etc.). In some embodiments, the circuit boards may include an input/output path.

(53) Each one of device **600** and device **601** may receive content and data via input/output (I/O) path **602**. I/O path **602** may provide content (e.g., broadcast programming, on-demand programming, Internet content, content available over a local area network (LAN) or wide area network (WAN), and/or other content) and data (e.g., training input and output images, data defining a pattern of a mask, and data defining a layout of portions of a display layer that are at least partially transparent) to control circuitry **604**, which may comprise processing circuitry **606** and storage **608**. Control circuitry **604** may be used to send and receive commands, requests, and other suitable data using I/O path **602**, which may comprise I/O circuitry. I/O path **602** may connect control circuitry **604** (and specifically processing circuitry **606**) to one or more communications paths (described below). I/O functions may be provided by one or more of these communications paths, but are shown as a single path in FIG. 6 to avoid overcomplicating the drawing. While set-top box **615** is shown in FIG. 7 for illustration, any suitable computing device having processing circuitry, control circuitry, and storage may be used in accordance with the present disclosure. For example, set-top box **615** may be replaced by, or complemented by, a personal computer (e.g., a notebook, a laptop, a desktop), a smartphone (e.g., device **600**), a tablet, a network-based server hosting a user-accessible client device, a non-user-owned device, any other suitable device, or any combination thereof.

(54) Control circuitry **604** may be based on any suitable control circuitry such as processing circuitry **606**. As referred to herein, control circuitry should be understood to mean circuitry based on one or more microprocessors, microcontrollers, digital signal processors, programmable logic devices, field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), etc., and may include a multi-core processor (e.g., dual-core, quad-core, hexa-core, or any suitable number of cores) or supercomputer. In some embodiments, control circuitry may be distributed across multiple separate processors or processing units, for example, multiple of the same type of processing units (e.g., two Intel Core i7 processors) or multiple different processors (e.g., an Intel Core i5 processor and an Intel Core i7 processor). In some embodiments, control circuitry **604** executes instructions for an image reconstruction application stored in memory (e.g., storage **608**). Specifically, control circuitry **604** may be instructed by the image reconstruction application to perform the functions discussed above and below. In some implementations, processing or actions performed by control circuitry **604** may be based on instructions received from the image reconstruction application.

(55) In client/server-based embodiments, control circuitry **604** may include communications circuitry suitable for communicating with a server or other networks or servers. The image reconstruction application may be a stand-alone application implemented on a device or a server. The image reconstruction application may be implemented as software or a set of executable instructions. The instructions for performing any of the embodiments discussed herein of the image reconstruction application may be encoded on non-transitory computer-readable media (e.g., a hard drive, random-access memory on a DRAM integrated circuit, read-only memory on a BLU-RAY disk, etc.). For example, in FIG. 6, the instructions may be stored in storage **608**, and executed by control circuitry **604** of a device **600**.

(56) In some embodiments, the image reconstruction application may be a client/server application where only the client application resides on device **600** (e.g., device **101** in FIG. 1), and a server application resides on an external server (e.g., server **704** and/or server of an edge computing device **716** below in FIG. 7). For example, the image reconstruction application may be

implemented partially as a client application on control circuitry **604** of device **600** and partially on server (e.g., server **704** below in FIG. 7) as a server application running on control circuitry (e.g., control circuitry **711** below in FIG. 7). Server **704** may be a part of a local area network with one or more of devices **600** or may be part of a cloud computing environment accessed via the internet. In a cloud computing environment, various types of computing services for performing searches on the internet or informational databases, providing image reconstruction capabilities, providing storage (e.g., for a database) or parsing data (e.g., using machine learning algorithms described above and below) are provided by a collection of network-accessible computing and storage resources (e.g., server **704** and/or edge computing device **716**), referred to as “the cloud.” Device **600** may be a cloud client that relies on the cloud computing capabilities from server **704** to determine whether processing (e.g., at least a portion of virtual background processing and/or at least a portion of other processing tasks) should be offloaded from the mobile device, and facilitate such offloading. When executed by control circuitry of server **704** or edge computing device **716**, the image reconstruction application may instruct control circuitry (e.g., control circuitry **711** or **718** below in FIG. 7) to perform processing tasks for the client device and facilitate the image reconstruction.

(57) Control circuitry **604** may include communications circuitry suitable for communicating with a server, edge computing systems and devices, a table or database server, or other networks or servers. The instructions for carrying out the above-mentioned functionality may be stored on a server (which is described in more detail below in connection with FIG. 7). Communications circuitry may include a cable modem, an integrated services digital network (ISDN) modem, a digital subscriber line (DSL) modem, a telephone modem, Ethernet card, or a wireless modem for communications with other equipment, or any other suitable communications circuitry. Such communications may involve the Internet or any other suitable communication networks or paths (which is described in more detail below in connection with FIG. 7). In addition, communications circuitry may include circuitry that enables peer-to-peer communication of user equipment devices, or communication of user equipment devices in locations remote from each other (described in more detail below).

(58) Memory may be an electronic storage device provided as storage **608** that is part of control circuitry **604**. As referred to herein, the phrase “electronic storage device” or “storage device” should be understood to mean any device for storing electronic data, computer software, or firmware, such as random-access memory, read-only memory, hard drives, optical drives, digital video disc (DVD) recorders, compact disc (CD) recorders, BLU-RAY disc (BD) recorders, BLU-RAY 3D disc recorders, digital video recorders (DVR, sometimes called a personal video recorder, or PVR), solid state devices, quantum storage devices, gaming consoles, gaming media, or any other suitable fixed or removable storage devices, and/or any combination of the same. Storage **608** may be used to store various types of content described herein as well as image reconstruction application data described above. Nonvolatile memory may also be used (e.g., to launch a boot-up routine and other instructions). Cloud-based storage, described below in relation to FIG. 7, may be used to supplement storage **608** or instead of storage **608**.

(59) Control circuitry **604** may include video generating circuitry and tuning circuitry, such as one or more analog tuners, one or more MPEG-2 decoders or other digital decoding circuitry, high-definition tuners, or any other suitable tuning or video circuits or combinations of such circuits. Encoding circuitry (e.g., for converting over-the-air, analog, or digital signals to MPEG signals for storage) may also be provided. Control circuitry **604** may also include scaler circuitry for upconverting and downconverting content into the preferred output format of device **600**. Control circuitry **604** may also include digital-to-analog converter circuitry and analog-to-digital converter circuitry for converting between digital and analog signals. The tuning and encoding circuitry may be used by device **600**, **601** to receive and to display, to play, or to record content. The tuning and encoding circuitry may also be used to receive image or video image reconstruction data. The

circuitry described herein, including for example, the tuning, image or video generating, encoding, decoding, encrypting, decrypting, scaler, and analog/digital circuitry, may be implemented using software running on one or more general purpose or specialized processors. Multiple tuners may be provided to handle simultaneous tuning functions (e.g., watch and record functions, picture-in-picture (PIP) functions, multiple-tuner recording, etc.). If storage **608** is provided as a separate device from device **600**, the tuning and encoding circuitry (including multiple tuners) may be associated with storage **608**.

(60) Control circuitry **604** may receive instruction from a user by way of user input interface **610**. User input interface **610** may be any suitable user interface, such as a remote control, mouse, trackball, keypad, keyboard, touch screen, touchpad, stylus input, joystick, voice recognition interface, or other user input interfaces. Display **612** may be provided as a stand-alone device or integrated with other elements of each one of device **600** and device **601**. For example, display **612** may be a touchscreen or touch-sensitive display. In such circumstances, user input interface **610** may be integrated with or combined with display **612**. In some embodiments, user input interface **610** includes a remote-control device having one or more microphones, buttons, keypads, any other components configured to receive user input or combinations thereof. For example, user input interface **610** may include a handheld remote-control device having an alphanumeric keypad and option buttons. In a further example, user input interface **610** may include a handheld remote-control device having a microphone and control circuitry configured to receive and identify voice commands and transmit information to set-top box **615**.

(61) Audio output equipment **614** may be integrated with or combined with display **612**. Display **612** may be one or more of a monitor, a television, an LCD for a mobile device, amorphous silicon display, low-temperature polysilicon display, electronic ink display, electrophoretic display, active matrix display, electro-wetting display, electro-fluidic display, cathode ray tube display, light-emitting diode display, electroluminescent display, plasma display panel, high-performance addressing display, thin-film transistor display, organic light-emitting diode display, surface-conduction electron-emitter display (SED), laser television, carbon nanotubes, quantum dot display, interferometric modulator display, or any other suitable equipment for displaying visual images. A video card or graphics card may generate the output to the display **612**. Audio output equipment **614** may be provided as integrated with other elements of each one of device **600** and device **601** or may be stand-alone units. An audio component of videos and other content displayed on display **612** may be played through speakers (or headphones) of audio output equipment **614**. In some embodiments, audio may be distributed to a receiver (not shown), which processes and outputs the audio via speakers of audio output equipment **614**. In some embodiments, for example, control circuitry **604** is configured to provide audio cues to a user, or other audio feedback to a user, using speakers of audio output equipment **614**. There may be a separate microphone **616** or audio output equipment **614** may include a microphone configured to receive audio input such as voice commands or speech. For example, a user may speak letters or words that are received by the microphone and converted to text by control circuitry **604**. In a further example, a user may voice commands that are received by a microphone and recognized by control circuitry **604**. Camera **620** may be any suitable image or video camera integrated with the equipment or externally connected. Camera **620** may be a digital camera comprising a CCD and/or a CMOS image sensor. In particular, camera **620** may include a lensless CMOS sensor masked with a mask (e.g., image sensor **108** and mask **110** in FIG. 1). Camera **620** may be an analog camera that converts to digital images via a video card.

(62) The image reconstruction application may be implemented using any suitable architecture. For example, it may be a stand-alone application wholly-implemented on each one of device **600** and device **601**. In such an approach, instructions of the application may be stored locally (e.g., in storage **608**), and data for use by the application is downloaded on a periodic basis (e.g., from an out-of-band feed, from an Internet resource, or using another suitable approach). Control circuitry

604 may retrieve instructions of the application from storage **608** and process the instructions to provide image reconstruction functionality and preform any of the actions discussed herein. Based on the processed instructions, control circuitry **604** may determine what action to perform when input is received from user input interface **610**. For example, movement of a cursor on a display up/down may be indicated by the processed instructions when user input interface **610** indicates that an up/down button was selected. An application and/or any instructions for performing any of the embodiments discussed herein may be encoded on computer-readable media. Computer-readable media includes any media capable of storing data. The computer-readable media may be non-transitory including, but not limited to, volatile and non-volatile computer memory or storage devices such as a hard disk, floppy disk, USB drive, DVD, CD, media card, register memory, processor cache, Random Access Memory (RAM), etc.

(63) In some embodiments, the image reconstruction application is a client/server-based application. Data for use by a thick or thin client implemented on each one of device **600** and device **601** may be retrieved on-demand by issuing requests to a server remote to each one of device **600** and device **601**. For example, the remote server may store the instructions for the application in a storage device. The remote server may process the stored instructions using circuitry (e.g., control circuitry **604**) and generate the displays discussed above and below. The client device may receive the displays generated by the remote server and may display the content of the displays locally on device **600**. This way, the processing of the instructions is performed remotely by the server while the resulting displays (e.g., that may include text, a keyboard, or other visuals) are provided locally on device **600**. Device **600** may receive inputs from the user via input interface **610** and transmit those inputs to the remote server for processing and generating the corresponding displays. For example, device **600** may transmit a communication to the remote server indicating that an up/down button was selected via input interface **610**. The remote server may process instructions in accordance with that input and generate a display of the application corresponding to the input (e.g., a display that moves a cursor up/down). The generated display is then transmitted to device **600** for presentation to the user.

(64) In some embodiments, the image reconstruction application may be downloaded and interpreted or otherwise run by an interpreter or virtual machine (run by control circuitry **604**). In some embodiments, the image reconstruction application may be encoded in the ETV Binary Interchange Format (EBIF), received by control circuitry **604** as part of a suitable feed, and interpreted by a user agent running on control circuitry **604**. For example, the image reconstruction application may be an EBIF application. In some embodiments, the image reconstruction application may be defined by a series of JAVA-based files that are received and run by a local virtual machine or other suitable middleware executed by control circuitry **604**. In some of such embodiments (e.g., those employing MPEG-2 or other digital media encoding schemes), image reconstruction application may be, for example, encoded and transmitted in an MPEG-2 object carousel with the MPEG audio and video packets of a program.

(65) FIG. 7 is a diagram of an illustrative system **700** for under-display image capture and image reconstruction, in accordance with some embodiments of this disclosure. User equipment devices **707**, **708**, **710** (e.g., which may correspond to one or more of device **101** may be coupled to communication network **706**). Communication network **706** may be one or more networks including the Internet, a mobile phone network, mobile voice or data network (e.g., a 5G, 4G, or LTE network), cable network, public switched telephone network, or other types of communication network or combinations of communication networks. Paths (e.g., depicted as arrows connecting the respective devices to the communication network **706**) may separately or together include one or more communications paths, such as a satellite path, a fiber-optic path, a cable path, a path that supports Internet communications (e.g., IPTV), free-space connections (e.g., for broadcast or other wireless signals), or any other suitable wired or wireless communications path or combination of such paths. Communications with the client devices may be provided by one or more of these

communications paths but are shown as a single path in FIG. 7 to avoid overcomplicating the drawing.

(66) Although communications paths are not drawn between user equipment devices, these devices may communicate directly with each other via communications paths as well as other short-range, point-to-point communications paths, such as USB cables, IEEE 1394 cables, wireless paths (e.g., Bluetooth, infrared, IEEE 702-11x, etc.), or other short-range communication via wired or wireless paths. The user equipment devices may also communicate with each other directly through an indirect path via communication network **706**.

(67) System **700** may comprise media content source **702**, one or more servers **704**, and one or more edge computing devices **716** (e.g., included as part of an edge computing system, such as, for example, managed by mobile operator). In some embodiments, the image reconstruction application may be executed at one or more of control circuitry **711** of server **704** (and/or control circuitry of user equipment devices **707**, **708**, **710** and/or control circuitry **718** of edge computing device **716**). In some embodiments, a data structure containing the training input and output images, may be stored at database **705** maintained at or otherwise associated with server **704**, and/or at storage **722** and/or at storage of one or more of user equipment devices **707**, **708**, **710**.

(68) In some embodiments, server **704** may include control circuitry **711** and storage **714** (e.g., RAM, ROM, Hard Disk, Removable Disk, etc.). Storage **714** may store one or more databases. Server **704** may also include an input/output path **712**. I/O path **712** may provide image reconstruction data, device information, or other data, over a local area network (LAN) or wide area network (WAN), and/or other content and data to control circuitry **711**, which may include processing circuitry, and storage **714**. Control circuitry **711** may be used to send and receive commands, requests, and other suitable data using I/O path **712**, which may comprise I/O circuitry. I/O path **712** may connect control circuitry **711** (and specifically control circuitry) to one or more communications paths.

(69) Control circuitry **711** may be based on any suitable control circuitry such as one or more microprocessors, microcontrollers, digital signal processors, programmable logic devices, field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), etc., and may include a multi-core processor (e.g., dual-core, quad-core, hexa-core, or any suitable number of cores) or supercomputer. In some embodiments, control circuitry **711** may be distributed across multiple separate processors or processing units, for example, multiple of the same type of processing units (e.g., two Intel Core i7 processors) or multiple different processors (e.g., an Intel Core i5 processor and an Intel Core i7 processor). In some embodiments, control circuitry **711** executes instructions for an emulation system application stored in memory (e.g., the storage **714**). Memory may be an electronic storage device provided as storage **714** that is part of control circuitry **711**.

(70) Edge computing device **716** may comprise control circuitry **718**, I/O path **720** and storage **722**, which may be implemented in a similar manner as control circuitry **711**, I/O path **712** and storage **722**, respectively of server **704**. Edge computing device **716** may be configured to be in communication with one or more of user equipment devices **707**, **708**, **710** and server **704**, which may be a video server, over communication network **706**, and may be configured to perform processing tasks (e.g., image reconstruction) in connection with ongoing processing of image or video data. In some embodiments, a plurality of edge computing devices **716** may be strategically located at various geographic locations, and may be mobile edge computing devices configured to provide processing support for mobile devices at various geographical regions.

(71) FIGS. **8A-8C** show illustrative diagrams of adjusting a pattern of a mask **810**, in accordance with some embodiments of this disclosure. In particular, each of FIGS. **8A-8C** show the mask **110** on the left side and an overhead view of a portion of the display layer **102** positioned over the mask **810** on the right side. The mask **810** includes portions **812** that are at least partially transparent and opaque material **814**. In the depicted embodiments, the mask **810** includes a transparent substrate

and the portions **812** are formed by transparent portions of the mask **810** substrate. The transparency of the mask **810** may be turned on and off, e.g., by control circuitry (e.g., control circuitry in FIG. **1**), to adjust a pattern of the mask **810** for allowing light that is directed through the portions **106** of the display layer **102** to be directed to an underlying image sensor (e.g., image sensor **108** in FIG. **1**). The transparency may be turned on and off by reconfiguring the opaque material **814**, which in turn reconfigures the portions **812**. In some embodiments, the opaque material **814** of the mask **810** may be turned on and off or reconfigured. In some embodiments, the mask **810** includes an electronic ink layer, which may include the opaque material **814**. In some embodiments, the opaque material **814** includes liquid crystals. The liquid crystals may be configured in different segments that may turn opaque when an electric charge is applied (e.g., the liquid crystals are turned on) and transparent when there is no electric charge (e.g., the liquid crystals are turned off), such as in a segmented LCD.

(72) Collectively, FIGS. **8A-8C** illustrate different patterns of the mask **810** resulting from turning on and off the transparency of the mask **810**. FIG. **8A** shows a first pattern of the mask **810** where a first subset of the portions **106** of the display layer **102** align with the portions **812** of the mask **810**. FIG. **8B** shows a second pattern of the mask **810** where the mask **810** is opaque. For example, the opaque material **814** of the mask **810** is arranged to block the light directed through the portions **106** of the display layer **102**. FIG. **8C** shows a third pattern of the mask **810** where a second subset of the portions **106** of the display layer **102** align with the portions **812** of the mask **810**. The second subset is different than the first subset in FIG. **8A**. The first and second subsets may include some of the same portions **812**.

(73) FIGS. **9A** and **9B** show illustrative diagrams of arrays of pixel elements (e.g., pixel elements **904** and **964**), in accordance with some embodiments of this disclosure.

(74) FIG. **9A** shows a display layer **902**, which includes the pixel elements **904** and portions **906** that are at least partially transparent. Each pixel element **904** includes subpixels **920**. In the depicted embodiment, the subpixels **920** are shaped as rounded rectangles (e.g., squircles) and ovals, which are smaller than the rounded rectangles. Each pixel element **904** includes two rounded rectangle subpixels **920** (e.g., each one emitting a different color) and two oval subpixels **920** (e.g., emitting the same color, but different than the colors of the rounded rectangles). The rounded rectangle subpixels **920** are arranged in a checkerboard pattern and the oval subpixels are positioned at intersections of the checkerboard pattern. The portions **906** are formed between the subpixels **920** such that they surround the subpixels **920** when viewed from above.

(75) FIG. **9B** shows a display layer **962**, which includes the pixel elements **964** and portions **966** that are at least partially transparent. In the depicted embodiment, each pixel element **964** includes four rectangular-shaped subpixels **970**. Two of the subpixels **970** (e.g., thin subpixels) emit the same color and have a length that is at least twice the width. The remaining two subpixels **970** (e.g., full subpixels) emit colors different from one another and different from the thin subpixels and have a length that is less than twice the width. The subpixels **970** are arranged in alternating columns, where one column contains the skinny subpixels and the other column contains the full subpixels alternating between the emitted color. The subpixels **970** do not touch another and the portions **966** are formed therebetween.

(76) Although FIGS. **9A** and **9B** show two pixel element configurations, other configurations may be used with the concepts disclosed herein.

(77) The embodiments discussed above are intended to be illustrative and not limiting. One skilled in the art would appreciate that individual aspects of the apparatus and methods discussed herein may be omitted, modified, combined, and/or rearranged without departing from the scope of the disclosure. Only the claims that follow are meant to set bounds as to what the present disclosure includes.

Claims

1. A system comprising: a display layer comprising: a plurality of pixel elements; and portions between the pixel elements that are at least partially transparent; an image sensor positioned under the display layer; a mask disposed between the image sensor and the display layer, wherein the mask comprises a configurable pattern for allowing light that is directed through a subset of the portions between the pixel elements to be directed to the image sensor; and control circuitry configured to change the pattern of the mask for allowing light by selectively turning on and off transparency of the mask.
2. The system of claim 1, wherein the portions between the pixel elements that are at least partially transparent comprise holes.
3. The system of claim 1, wherein the portions between the pixel elements that are at least partially transparent comprise a transparent material.
4. The system of claim 1, further comprising a lensless camera, wherein the lensless camera comprises the image sensor, the mask, and a portion of the display layer disposed above the mask.
5. The system of claim 1, wherein the control circuitry is configured to: set the mask to be opaque when a subset of the plurality of pixel elements emit light and the image sensor is not active, the subset of the plurality of pixel elements disposed above the mask; and set at least a portion of the mask to be transparent when the subset of the plurality of pixel elements do not emit light and the image sensor is active.
6. The system of claim 1, wherein: the mask comprises an electronic ink layer; and the control circuitry is configured to turn on and off the transparency of the mask by reconfiguring the electronic ink layer.
7. The system of claim 1, wherein the image sensor is configured to generate a captured image based on the light that is directed through the subset of the portions between the pixel elements and through the pattern of the mask.
8. The system of claim 7, further comprising control circuitry configured to generate a reconstructed image based on the captured image and data defining the pattern of the mask.
9. The system of claim 8, wherein the control circuitry is configured to generate the reconstructed image by inputting the captured image into a trained machine learning model, wherein the machine learning model was trained by: accessing a training input image and a corresponding training output image, wherein the training input image is specific to the pattern of the mask; inputting the training input image into the machine learning model to generate an attempted reconstructed image; and adjusting the machine learning model based on comparing the attempted reconstructed image with the training output image.
10. The system of claim 1, wherein pixel elements of the plurality of pixel elements comprise a casing to block light emitted by the pixel elements from the image sensor.
11. The system of claim 1, further comprising a plurality of microlenses, wherein: the pattern of the mask allows light that is directed through a subset of the plurality of microlenses to be directed to the image sensor.
12. The system of claim 1, wherein the control circuitry is configured to set the mask to be opaque when the image sensor is not active.
13. The system of claim 1, wherein the control circuitry is configured to change the pattern of the mask from a first pattern to a second pattern based at least in part on a received command to modify privacy.
14. A method comprising: generating a reconstructed image based on a captured image and data defining a pattern of a mask, wherein: the captured image is based on a light that was directed through the pattern of the mask and through a display layer, the display layer comprising: a plurality of pixel elements; and portions between the pixel elements that are at least partially

transparent; and the pattern of the mask is configurable and allows at least a portion of light that is directed through a subset of the portions between the pixel elements to be directed through the pattern of the mask; and storing the reconstructed image in memory; and changing the pattern of the mask by selectively turning on and off transparency of the mask.

15. The method of claim 14, wherein the portions between the pixel elements that are at least partially transparent comprise holes.

16. The method of claim 14, wherein the portions between the pixel elements that are at least partially transparent comprise a transparent material.

17. The method of claim 14, wherein: the captured image is captured using a lensless camera; the lensless camera comprises an image sensor, the mask, and the subset of the portions between the pixel elements; and the light that is directed through the pattern of the mask is directed to the image sensor.

18. The method of claim 14, further comprising: setting the mask to be opaque when a subset of the plurality of pixel elements emit light, the subset of the plurality of pixel elements disposed above the mask; and setting at least a portion of the mask to be transparent when the subset of the plurality of pixel elements do not emit light.

19. The method of claim 14, further comprising inputting the captured image into a trained machine learning model, wherein the machine learning model was trained by: accessing a training input image and a corresponding training output image, wherein the training input image is specific to the pattern of the mask; inputting the training input image into the machine learning model to generate an attempted reconstructed image; and adjusting the machine learning model based on comparing the attempted reconstructed image with the training output image.

20. The method of claim 14, further comprising displaying an image on the display layer, when capturing the captured image, that is different than the captured image and the reconstructed image.
